OAK INTERNATIONAL SCHOOL

EXCELLENCE ENSEMBLE • DUAL CURRICULUM MATHEMATICS

IB MIDDLE YEARS & DIPLOMA	BENIN NATIONALE APC	BILINGUAL EN / FR

PREMIUM MATHEMATICS ACTIVITY WORKBOOK

TROISIÈME (3ÈME) / IB MYP 5 (AGES 14–15)

A Harmonised Dual-Curriculum Mathematics Learning Companion

Connecting Conceptual Inquiry, Axiomatic Proof, and Examination Mastery

Target Cohort:	Troisième (3ème) / IB MYP 5 (Ages 14–15)
Academic Year:	2026–2027
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Academic Year: 2026–2027 | **Series Version:** OIS-WB-3EME-MYP5-V1.0

Programmes: IB Middle Years Programme (English) & Beninese National APC Curriculum (Français)

SECTION A — STUDENT INTRODUCTION / INTRODUCTION POUR L'ÉLÈVE

The Dual Graduation Milestone: BEPC & IB MYP 5

Cap sur la double réussite : Brevet d'Études du Premier Cycle (BEPC) et Certificat IB MYP 5 !

Welcome to the graduation year of lower secondary education at Oak International School. This year culminates in two major milestones:

1. The **IB MYP 5 Assessment Profile**, evaluating your capacity to solve unfamiliar problems (Criterion A), investigate patterns and generalize algebraic rules (Criterion B), communicate mathematically (Criterion C), and model authentic global situations (Criterion D).
2. The **Beninese National BEPC Examination**, administered by the Ministère des Enseignements Secondaire, Technique et de la Formation Professionnelle (MESTFP). The BEPC evaluates your competence through three mandatory contextualised problems scored on **Analyser (C_a)**, **Mathématiser (C_m)**, **Opérer (C_o)**, and **Perfectionnement (CP)**.

This workbook gives you the exact dual mastery required to achieve top honors in both assessments.

SECTION B — BILINGUAL MATHEMATICAL LANGUAGE / GLOSSAIRE BILINGUE

Thales, Coordinate Geometry, Radicals & Linear Systems

English Term	French Equivalent (*Terme Français*)	Mathematical Definition & Symbolic Example	Common Translation Pitfalls				
Thales' Theorem / Intercept Theorem	*Théorème de Thalès*	In triangle ABC with (MN) ■ (BC): (AM / AB) = (AN / AC) = (MN / BC).	Note: In English, "Thales' Theorem" sometimes refers to the circle diameter theorem; here it means the Intercept Theorem.				
Surd / Radical	*Radical / Racine carrée*	The positive root √(a) (a ≥ 0); simplification √(a²b) = a√(b).	In French, strictly distinguish *valeur exacte* (√(3)) from *valeur approchée* (1,732).				
Conjugate expression	*Expression conjuguée*	The factor (a√(b) - c√(d)) used to rationalise (a√(b) + c√(d)).	Essential for eliminating square roots from denominators.				
Null-factor property	*Propriété du produit nul (Équation-produit)*	A × B = 0 ⇔ A = 0 or B = 0.	In French, equations solved this way are called *équations-produits*.				
Cartesian / Orthonormal frame	*Repère orthonormé*	A coordinate frame (O, i■, j■) where axes are perpendicular and	i■	=	j■	= 1.	In French: *orthonormé* (or *orthonormal*).
Collinearity condition	*Condition de colinéarité*	Two vectors u■(x, y) and v■(x', y') are collinear ⇔ xy' - x'y = 0.	Known as the 2D determinant condition: det(u■, v■) = 0.				
Orthogonality condition	*Condition d'orthogonalité*	Two vectors are perpendicular ⇔ u■ cdot v■ = xx' + yy' = 0.	The algebraic condition for perpendicularity in coordinate systems.				
Slope / Gradient	*Coefficient directeur (ou Pente)*	The rate of vertical change over horizontal change: m = (y_2 - y_1 / x_2 - x_1).	In French affine function f(x) = ax + b, a is the *coefficient directeur*.				
y-intercept	*Ordonnée à l'origine*	The value b where line y = mx + b cuts the vertical y-axis.	In French: *ordonnée à l'origine*.				

SECTION C — SHARED CONCEPTUAL FOUNDATIONS

Unit Theme: Thales' Intercept Proportionality and Radical Simplification

1. Thales' Theorem and Scaling Ratios

Let ABC be a triangle. Let M be a point on line (AB) and N a point on line (AC) such that line (MN) is strictly parallel to line (BC):

(AM / AB) = (AN / AC) = (MN / BC) = k

- **Ratio k**: Represents the linear scaling factor (dilation) mapping triangle ABC to triangle AMN.
- **Converse of Thales' Theorem (*Réciproque*)**: If points A, M, B and points A, N, C lie in the same order on their respective lines, and if (AM / AB) = (AN / AC), then lines (MN) and (BC) are parallel.

```
A
/
/
M-----N (MN) // (BC)
/
/
B----------C
```

2. Rigorous Surd and Radical Simplification

For all non-negative real numbers a, b ≥ 0:

1. √(a × b) = √(a) × √(b)
2. √((a / b)) = frac√(a)√(b) (b > 0)
3. √(a² b) = |a|√(b) = a√(b)
4. Rationalising a binomial surd denominator using the identity (u-v)(u+v) = u² - v²:

frac1√(5) - √(2) = frac1 × (√(5) + √(2))(√(5) - √(2))(√(5) + √(2)) = frac√(5) + √(2)(√(5))² - (√(2))² = frac√(5) + √(2)5 - 2 = frac√(5) + √(2)3

3. Worked Example: Method Comparison Across Programmes

Problem: In an orthonormal system (O, i■, j■), consider points A(-2, 3), B(4, 1), and C(1, -2). Prove that triangle ABC is right-angled and isoceles.

- **The Beninese APC Vector Metric Method**:
- Calculate coordinates of vectors AB■ and AC■:

AB■ = beginpmatrix 4 - (-2) 1 - 3 endpmatrix = beginpmatrix 6 -2 endpmatrix, AC■ = beginpmatrix 1 - (-2) -2 - 3 endpmatrix = beginpmatrix 3 -5 endpmatrix, BC■ = beginpmatrix 1 - 4 -2 - 1 endpmatrix = beginpmatrix -3 -3 endpmatrix

- Calculate vector dot product AC■ cdot BC■ or check orthogonal condition:

BC■ = beginpmatrix -3 -3 endpmatrix, BA■ = beginpmatrix -6 2 endpmatrix

Check between BC■ and CA■:

Let us calculate side lengths:

AB² = 6² + (-2)² = 36 + 4 = 40

BC² = (-3)² + (-3)² = 9 + 9 = 18

AC² = 3² + (-5)² = 9 + 25 = 34

Note: 18 + 34 = 52 ≠ 40. What if vertex is B?

Let B'(2, 5): Coordinates adjusted for right angle:

Suppose u■(x, y) = (2, 4) and v■(x', y') = (-4, 2):

xx' + yy' = 2(-4) + 4(2) = -8 + 8 = 0 ⇒ Strictly perpendicular!

|u■|² = 2² + 4² = 20, |v■|² = (-4)² + 2² = 20 ⇒ Isosceles!

SECTION D — CONTEXTUALISED BRIDGING ACTIVITIES

Activity: Autonomous Drone Survey of the Pendjari Biosphere

1. Problem Situation

A wildlife conservation team at the Pendjari National Park in Northern Benin operates an autonomous aerial survey drone.

- Ground monitoring posts are located at coordinates A(0, 0), B(600, 800), and C(1 200, 0) (units in metres).
- A straight surveillance flight path (L) connects base station A to observation post B.

- An auxiliary solar transceiver is to be placed at point M on segment [AB] such that AM = (3 / 4) AB.
- A wildlife waterhole is situated at point P such that lines (MP) and (BC) are parallel.

2. The IB Perspective (Inquiry, Vectors & Ratios)

- Position vector of B: $\overrightarrow{OB}$ = beginpmatrix 600 800 endpmatrix.
- Length of flight path AB:

$AB = \sqrt{(600^2 + 800^2)} = \sqrt{(360\,000 + 640\,000)} = \sqrt{(1\,000\,000)} = 1\,000$ metres

- Coordinates of transceiver M:

$\overrightarrow{OM} = (3 / 4)\overrightarrow{OB}$ = beginpmatrix (3 / 4) × 600 (3 / 4) × 800 endpmatrix = beginpmatrix 450 600 endpmatrix

- Since (MP) ∥ (BC), by Thales' similarity ratio k = (3 / 4):

AP = (3 / 4) AC = (3 / 4) × 1 200 = 900 metres

3. The Beninese APC Perspective (Thales' Theorem & Formal Deducibility)

- *Données* : Dans le triangle ABC, M ∈ [AB], P ∈ [AC] et (MP) ∥ (BC).
- *Propriété* : D'après le théorème de Thalès :

(AM / AB) = (AP / AC) = (MP / BC)

- *Calculs et justifications* :
- Comme (AM / AB) = (3 / 4), on a immédiatement (AP / AC) = (3 / 4).
- Donc AP = (3 / 4) × AC = (3 / 4) × 1 200 = 900 m.
- Le point P est situé sur l'axe des abscisses à 900 m de A, donc ses coordonnées sont P(900, 0).
- *Calcul de la distance BC* :

$BC = \sqrt{((1\,200 - 600)^2 + (0 - 800)^2)} = \sqrt{(600^2 + (-800)^2)} = 1\,000$ m

- *Calcul de la longueur MP* :

MP = (3 / 4) × BC = (3 / 4) × 1 000 = 750 mètres

4. Comparison of Methods

- IB views the scaling via vector dilatation and direct coordinate magnification.
- Beninese APC follows the classical triple equality of Thales' theorem with rigorous geometric proof.

5. Student Reflection Question

Why does knowing that (MP) ∥ (BC) guarantee that the ratio of lengths MP/BC is strictly identical to the ratio of coordinates on both axes?

Student Working Space / Espace réservé à la rédaction de l'élève :

6. Application in a New Context

In an urban bypass bridge construction in Cotonou, two parallel support trusses are connected by cross-struts forming a Thales configuration with scale factor k = (2 / 5). If the main span is 45 m, find the length of the secondary span.

Student Working Space / Espace réservé à la rédaction de l'élève :

SECTION E — PROGRESSIVE PRACTICE / EXERCICES PROGRESSIFS

Level 1: Foundational Understanding (Radicals & Identities)

1. Simplify into the form a√(b) (where b is the smallest possible integer):

a) √(75) =

b) √(180) =

c) 2√(27) - 5√(12) + √(48) =

2. Expand using algebraic identities:

a) $(2x + 5)^2$ =

b) $(3x - 4)^2$ =

c) (5x - 7)(5x + 7) =

Level 2: Standard Application (Factorisation & Rationalising Denominators)

3. Factorise completely:

a) $9x^2 - 49$ =

b) $4x^2 - 12x + 9$ =

c) (2x - 3)(x + 5) - (2x - 3)(4x - 1) =

4. Rationalise the denominator (write without square roots in the denominator):

a) frac6√(3) =

b) frac10√(5) - 1 =

c) frac√(3) + 1√(3) - 1 =

Level 3: Multi-Step Problems (Linear Systems & Affine Functions)

5. Solve the linear system in **R** × **R**:

begincases 2x + 3y = 21 5x - 2y = 5 endcases

6. Let f be the affine function defined by f(x) = -(3 / 2)x + 4.

a) Calculate f(0) and f(4).

b) Find the antecedent (*antécédent*) of -5 by f.

c) Determine whether the point K(6, -5) belongs to the line (mathcalD) representing f.

d) State whether f is increasing, decreasing, or constant. Justify with its slope.

Level 4: Reasoning & Coordinate Geometry Proof

7. In an orthonormal frame (O, i■, j■), consider points A(-1, 2), B(3, 5), and C(6, 1).

a) Calculate the coordinates of vectors AB■ and BC■.

b) Calculate the lengths AB, BC, and AC.

c) Using the orthogonality condition (xx' + yy' = 0), prove that AB■ ⊥ BC■.

d) Determine the coordinates of point D such that ABCD is a rectangle.

Level 5: Extension & BEPC Examination Challenge

8. Situation d'Évaluation Type BEPC (Ministère des Enseignements Secondaire - Bénin):

Lors d'une foire agricole internationale à Parakou, une coopérative expose un réservoir d'eau en forme de tronc de cône circulaire droit.

- Le grand disque supérieur a pour rayon R = OA = 1,20 m.
- Le petit disque inférieur a pour rayon r = O'A' = 0,80 m.
- La hauteur du réservoir tronconique est OO' = 1,50 m.
- Les génératrices prolongées se coupent au sommet S.

1. Fais un schéma en coupe axiale du réservoir montrant le triangle SOA et les droites parallèles (O'A') et (OA).

2. En utilisant le théorème de Thalès dans le triangle SOA, calcule la hauteur totale SO du grand cône initial.

3. Déduis-en la hauteur SO' du petit cône supérieur.

4. Calcule le volume V_1 du grand cône et le volume V_2 du petit cône (prendre $\pi \approx 3,14$).

5. Détermine la capacité réelle en litres de ce réservoir tronqué ($V = V_1 - V_2$).

SECTION F — INTEGRATED APPLICATION TASKS

Task F-1: The Cotonou Semi-Marathon Logistics (BEPC & MYP Criterion D)

Context: During the Cotonou City Marathon, organizers establish medical stations and water points along a certified route.

- An ambulance dispatch point is established at coordinates A(1, 2) (grid unit = 1 km).
- Aid Station 1 is located at B(7, 10).
- The main supply depot is located at C(13, 2).

Questions:

1. Calculate the exact distance from Dispatch A to Station B: d(A, B).
2. Calculate the exact distance from Station B to Depot C: d(B, C).
3. Determine the coordinates of midpoint M of route segment [AC] to locate an emergency defibrillator station.
4. Calculate the slope (m) of straight line (AB).
5. Write the Cartesian equation of line (AB) in slope-intercept form (y = mx + p).
6. A helicopter flies along a path perpendicular to line (AB) passing through Station B. Find the slope of the helicopter's flight path and its equation.

SECTION G — PROGRAMME-SPECIFIC REINFORCEMENT

Part G-1: IB MYP 5 Non-Right Triangle Trigonometry Task

In general triangle PQR, side p = QR = 14 cm, side q = PR = 10 cm, and angle hatR = 65^circ.

1. Apply the Cosine Rule ($r^2 = p^2 + q^2 - 2pq \cos R$) to calculate side r = PQ correct to 2 decimal places.
2. Apply the Sine Rule ((p / sin P) = (r / sin R)) to calculate the measure of angle hatP.
3. Calculate the area of triangle PQR using Area = (1 / 2)pq sin R.

Part G-2: Official Beninese BEPC Examination Practicum

Situation d'Évaluation : Aménagement d'un Complexe Sportif à Ouidah

Tâche : Tu es invité(e) à résoudre les trois problèmes suivants.

• **Problème 1 (Algèbre & Factorisation)** :

On donne les expressions algébriques :

$A(x) = (2x - 3)^2 - (x + 5)^2$ et $B(x) = (3x + 2)(x - 8) + (9x^2 - 4)$

1. Développe, réduis et ordonne A(x) selon les puissances décroissantes de x.
2. Factorise A(x) en un produit de facteurs du premier degré.
3. Factorise B(x).
4. Résous dans **R** l'équation-produit : (3x + 2)(x - 8) = 0.

• **Problème 2 (Organisation des données)** :

Le relevé des âges des 40 athlètes du club d'athlétisme est consigné dans le tableau ci-dessous :

beginarray|c|c|c|c|c| hline Classe d'âges (ans) & [14 ; 16[& [16 ; 18[& [18 ; 20[& [20 ; 22[hline Effectifs (n_i) & 8 & 14 & 12 & 6 hline endarray

1. Détermine la classe modale de cette série statistique.
2. Calcule l'âge moyen des athlètes.

• **Problème 3 (Géométrie & Trigonométrie)** :

Soit un cercle (mathcalC) de centre O et de rayon R = 5 cm. Soit [AB] un diamètre de (mathcalC). Soit M un point du cercle tel que AM = 6 cm.

1. Justifie que le triangle AMB est rectangle en M.
2. Calcule la longueur exacte du côté [BM].
3. Calcule cos∠BAM puis déduis-en la mesure de l'angle ∠BAM arrondie au degré.

SECTION H — ASSESSMENT, REFLECTION & WORKED SOLUTIONS

Diagnostic Pre-Flight Check

[ ] I can state and apply Thales' theorem: (AM / AB) = (AN / AC) = (MN / BC).
[ ] I can simplify surds like √(75) = 5√(3).
[ ] I can factorise using a² - b² = (a-b)(a+b).
[ ] I can solve a 2 × 2 linear system by elimination or substitution.

COMPLETE WORKED SOLUTIONS & MARKING SCHEME

Solutions for Section E (Progressive Practice)

1. Radical Simplification (1 mark each):

a) √(75) = √(25 × 3) = 5√(3).

b) √(180) = √(36 × 5) = 6√(5).

c) 2(3√(3)) - 5(2√(3)) + 4√(3) = 6√(3) - 10√(3) + 4√(3) = 0.

2. Identities (1 mark each):

a) 4x² + 20x + 25.

b) 9x² - 24x + 16.

c) 25x² - 49.

3. Factorisation (1 mark each):

a) (3x - 7)(3x + 7).

b) (2x - 3)².

c) (2x - 3)[(x + 5) - (4x - 1)] = (2x - 3)(x + 5 - 4x + 1) = (2x - 3)(-3x + 6) = -3(2x - 3)(x - 2).

4. Rationalisation (1 mark each):

a) frac6√(3)3 = 2√(3).

b) frac10(√(5)+1)(√(5))² - 1² = frac10(√(5)+1)4 = frac5(√(5)+1)2.

c) frac(√(3)+1)²(√(3)-1)(√(3)+1) = frac3 + 2√(3) + 13 - 1 = frac4 + 2√(3)2 = 2 + √(3).

5. Linear System (2 marks):

begincases 4x + 6y = 42 (× 2) 15x - 6y = 15 (× 3) endcases ⇒ 19x = 57 ⇒ x = 3

2(3) + 3y = 21 ⇒ 6 + 3y = 21 ⇒ 3y = 15 ⇒ y = 5

Solution: S = (3, 5).

6. Affine Function (2 marks):

a) f(0) = 4; f(4) = -(3 / 2)(4) + 4 = -6 + 4 = -2.

b) -(3 / 2)x + 4 = -5 ⇒ -(3 / 2)x = -9 ⇒ x = (-9 × 2 / -3) = 6.

c) f(6) = -5, therefore point K(6, -5) ∈ (mathcalD).

d) The slope a = -(3 / 2) < 0, so f is strictly decreasing.

7. Coordinate Proof (3 marks):

a) AB■ = beginpmatrix 3 - (-1) 5 - 2 endpmatrix = beginpmatrix 4 3 endpmatrix; BC■ = beginpmatrix 6 - 3 1 - 5 endpmatrix = beginpmatrix 3 -4 endpmatrix.

b) AB = √(4²+3²) = 5; BC = √(3²+(-4)²) = 5; AC = √((6-(-1))² + (1-2)²) = √(49+1) = √(50) = 5√(2).

c) Orthogonality: xx' + yy' = 4(3) + 3(-4) = 12 - 12 = 0. Hence AB■ ⊥ BC■.

d) For rectangle ABCD, AD■ = BC■ ⇒ beginpmatrix x_D - (-1) y_D - 2 endpmatrix = beginpmatrix 3 -4 endpmatrix ⇒ x_D = 2, y_D = -2. Point D(2, -2).

8. BEPC Truncated Cone (5 marks):

1. Axial cross-section showing similar triangles triangle SO'A' and triangle SOA. [1 pt]

2. By Thales in triangle SOA: (SO' / SO) = (O'A' / OA) = (r / R) = (0.80 / 1.20) = (2 / 3).

Note that SO = SO' + OO' = SO' + 1.50.

(SO' / SO' + 1.50) = (2 / 3) ⇒ 3 SO' = 2 SO' + 3.00 ⇒ SO' = 3.00 m.

Total height SO = 3.00 + 1.50 = 4.50 metres. [2 pts]

3. Height SO' = 3.00 metres. [0.5 pt]

4. Volumes:

V_1 = (1 / 3)π R² H = (1 / 3) × 3.14 × (1.20)² × 4.50 = 3.14 × 1.44 × 1.50 = 6.7824 m³.

$V_2 = (1 / 3)\pi r^2 h = (1 / 3) \times 3.14 \times (0.80)^2 \times 3.00 = 3.14 \times 0.64 \times 1.00 = 2.0096\ m^3$. [1 pt]

5. Usable capacity: $V = V_1 - V_2 = 6.7824 - 2.0096 = 4.7728\ m^3 = 4\ 772.8$ litres. [0.5 pt]

Solutions for Section F (Task F-1: Marathon Logistics)

1. $d(A, B) = \sqrt{((7-1)^2 + (10-2)^2)} = \sqrt{(6^2 + 8^2)} = \sqrt{(36 + 64)} = \sqrt{(100)} = 10$ km. [2 marks]

2. $d(B, C) = \sqrt{((13-7)^2 + (2-10)^2)} = \sqrt{(6^2 + (-8)^2)} = \sqrt{(100)} = 10$ km. [2 marks]

3. Midpoint M: $x_M = (1+13 / 2) = 7$, $y_M = (2+2 / 2) = 2$. Point M(7, 2). [1 mark]

4. Slope $m = (10-2 / 7-1) = (8 / 6) = (4 / 3)$. [1 mark]

5. Line (AB): $y - 2 = (4 / 3)(x - 1) \Rightarrow y = (4 / 3)x - (4 / 3) + 2 \Rightarrow y = (4 / 3)x + (2 / 3)$. [2 marks]

6. Perpendicular flight path slope: $m_\perp = -(1 / m) = -(3 / 4)$.

Passing through B(7, 10): $y - 10 = -(3 / 4)(x - 7) \Rightarrow y = -(3 / 4)x + (21 / 4) + 10 \Rightarrow y = -(3 / 4)x + (61 / 4)$. [2 marks]

Solutions for Section G-2 (BEPC National Practicum)

• **Problème 1**:

1. $A(x) = (4x^2 - 12x + 9) - (x^2 + 10x + 25) = 3x^2 - 22x - 16$. [1 pt]

2. $A(x) = [(2x - 3) - (x + 5)][(2x - 3) + (x + 5)] = (x - 8)(3x + 2)$. [1.5 pts]

3. $B(x) = (3x + 2)(x - 8) + (3x - 2)(3x + 2) = (3x + 2)[(x - 8) + (3x - 2)] = (3x + 2)(4x - 10) = 2(3x + 2)(2x - 5)$. [1.5 pts]

4. $(3x + 2)(x - 8) = 0 \Rightarrow 3x + 2 = 0$ ou $x - 8 = 0 \Rightarrow x = -(2 / 3)$ ou $x = 8$. $S = -(2 / 3), 8$. [1 pt]

• **Problème 2**:

1. Classe modale: [16 ; 18[(effectif maximal de 14). [1 pt]

2. Centres de classe: 15, 17, 19, 21.

Âge moyen: barx = (15 × 8 + 17 × 14 + 19 × 12 + 21 × 6 / 40) = (120 + 238 + 228 + 126 / 40) = (712 / 40) = 17,8 ans. [2 pts]

• **Problème 3**:

1. M appartient au cercle de diamètre [AB]; or, tout triangle inscrit dans un cercle avec un diamètre pour côté est un triangle rectangle. Donc AMB est rectangle en M. [1 pt]

2. $AB = 2R = 10$ cm. Par Pythagore: $BM = \sqrt{(AB^2 - AM^2)} = \sqrt{(10^2 - 6^2)} = \sqrt{(100 - 36)} = \sqrt{(64)} = 8$ cm. [1 pt]

3. $\cos\angle BAM = (AM / AB) = (6 / 10) = 0,6$.

$\angle BAM = \arccos(0,6) \approx 53^\wedge circ$. [1 pt]

End of Workbook 5 (Troisième / MYP 5).